

INSTALLATION GUIDE

App version: v0.1.0-dev

This guide must be updated after installation, infrastructure, dependency, model, database, backup or environment changes.

OVERVIEW

This application runs on an Ubuntu 24.04 computer with OpenClaw, PostgreSQL, Ollama qwen3:4b, FastAPI, React and background workers.

REQUIRED SERVICES

- OpenClaw
- PostgreSQL
- Ollama with qwen3:4b
- FastAPI backend
- React frontend
- Background workers

ENVIRONMENT VARIABLES

Use .env.example as the template. Do not commit the real .env file.

POSTGRES SQL SETUP

Use Docker Compose or native PostgreSQL. PostgreSQL must not be exposed to the internet.

Stage 0 Docker Compose configuration is provided in the repository root:

```
bash
docker compose up -d postgres
```

The included configuration binds PostgreSQL to localhost only:

```
text
127.0.0.1:5432:5432
```

Required extensions for the first migration:

- pgcrypto
- uuid-ossdp

OLLAMA SETUP

Ollama must run locally only and the approved V1 local model is qwen3:4b.

Recommended secure defaults:

```
bash
export PATH="$HOME/.local/bin:$PATH" # only needed for a user-space install
export OLLAMA_HOST=127.0.0.1:11434
ollama serve
```

In another terminal, install and validate the approved model:

```
bash
./scripts/install_ollama_qwen3_4b.sh
./scripts/test_ollama_qwen3_4b.sh
```

The validation scripts fail if Ollama is unreachable, if qwen3:4b is missing, or if unexpected local Ollama models are present.

PAID MODEL SETUP

Configure ChatGPT as the primary development model and Anthropic as fallback.

Stage 0 model routing policy:

1. Ollama qwen3:4b for low-risk cleanup, classification, extraction, tagging and first-pass summaries.
2. ChatGPT/OpenAI as the primary paid development/reasoning provider.
3. Anthropic as fallback when ChatGPT/OpenAI token, context, rate or quota limits block progress.
4. Latest suitable paid models only for final higher-risk production/research analysis later.

Configure API keys only in the real .env file or host secret manager:

```
text
OPENAI_API_KEY=...
ANTHROPIC_API_KEY=...
```

Do not commit real API keys. Do not expose paid provider keys to frontend code.

Fallback events are recorded through the model_task_logs foundation. Logs must not store prompts, raw responses, API keys or secrets.

BACKUP AND RESTORE

Use the backup and restore scripts in /backup.

Stage 0 scripts:

- backup/backup_postgres.sh
- backup/restore_postgres.sh
- backup/backup_app_config.sh
- backup/restore_app_config.sh

Do not include unencrypted secrets in backups.

VALIDATION CHECKLIST

Run structural validation:

```
bash
./scripts/stage0_validate.sh
```

Stage 0 checklist:

- Backend skeleton imports and exposes /health
- Frontend skeleton exists and builds once Node dependencies are installed
- PostgreSQL Docker Compose config binds to localhost only
- Ollama is installed locally on 127.0.0.1:11434 and responds with qwen3:4b
- ChatGPT primary model is configured through environment/provider settings
- Anthropic fallback is configured through environment/provider settings
- Mock fallback tests cover token/rate-limit fallback behaviour
- model_task_logs migration foundation exists for fallback event logging
- ASVS register exists and records that latest stable ASVS must be checked at audit time
- No secrets are committed